MEMVLN: Streaming Episodic Memory for Vision-Language Navigation

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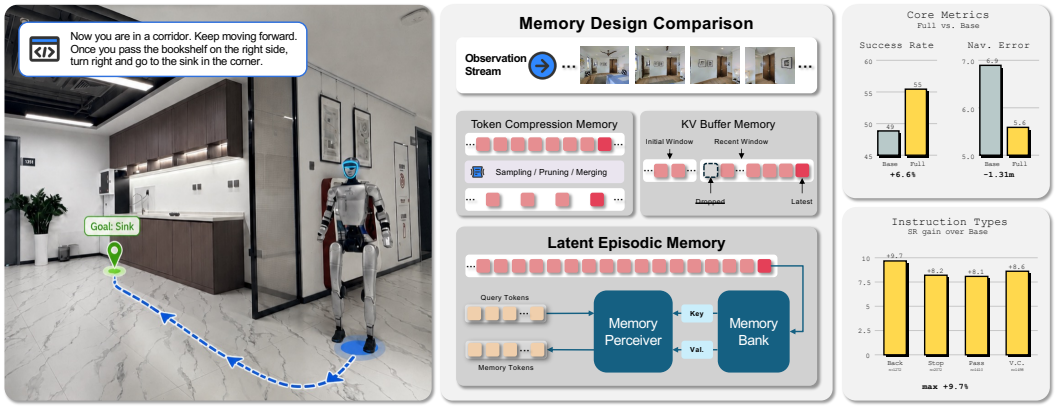


Figure 1: **MEMVLN overview.** We design a streaming episodic memory framework that endows vision-language models (VLMs) with navigation memory for streaming video. MEMVLN employs a memory module that progressively stores compressed perceptual features and produces compact memory tokens via cross-attention. MEMVLN elevates vision-language navigation performance, including long-horizon and real-world scenarios.

Abstract: Vision-language navigation (VLN) requires an embodied agent to follow language instructions through 3D environments. The prevailing formulation processes monocular streaming video with VLMs, yet existing methods are fundamentally constrained by the absence of persistent episodic memory, which is a critical limitation for long-horizon navigation in the real world. We present MEMVLN, a streaming episodic memory framework that endows VLMs with navigation memory for streaming video. MEMVLN introduces a memory bank that progressively stores compressed perceptual features and spatio-temporal information from the streaming video, and a memory perceiver that produces compact memory tokens via cross-attention. The memory module, together with a pretrained geometry encoder, integrates seamlessly into VLMs with optimized efficiency for streaming video. To learn the memory module, we design a masking-based trajectory summarization task with multi-granularity sub-trajectory decomposition, encouraging the model to reason exclusively from the episodic memory. Experiments on standard VLN-CE benchmarks demonstrate that MEMVLN achieves state-of-the-art performance when trained with comparable data. We further underscore the effectiveness of our episodic memory through long-horizon navigation analysis and real-world robot deployment. In addition, we conduct post-training with 24 demonstrations to further enhance real-world performance, demonstrating the adaptability of MEMVLN to real-world navigation.

Keywords: Vision-Language Navigation, Episodic Memory, Embodied AI

1 Introduction

Vision-language navigation (VLN) is a fundamental task in embodied AI, requiring an embodied agent to follow natural language instructions through 3D environments [1, 2, 3]. Recent advances in

Submitted to the 10th Conference on Robot Learning (CoRL 2026). Do not distribute.

vision-language models (VLMs) have substantially elevated performance in navigation tasks, with the prevailing formulation converging towards monocular streaming video input, a natural and practical interface for real-world deployment [4, 5, 6, 7]. Despite rapid progress, the absence of persistent memory remains a critical constraint for long-horizon navigation in the real world, where an agent relies on its full history of observations to make informed decisions. In this paper, we address the key problem: *how to design a simple yet effective episodic memory that integrates seamlessly with streaming VLM to elevate VLN capability*?

Navigation memory has received prior attention, yet existing approaches remain insufficient along several key dimensions. Early methods employ implicit recurrent modules [8, 9, 10], which lack the representational capacity to preserve the rich perceptual history required for long-horizon navigation. Explicit memory representations achieve stronger capacity, but are architecturally coupled to specific settings such as panoramic observations or topological maps, rendering them incompatible with streaming video in continuous environments [11, 12, 13, 14, 15]. More recent works revisit navigation memory within a streaming formulation [16, 17], yet the capacity limitation persists. These shortcomings collectively motivate our approach: *an episodic memory design that is expressive and natively compatible with streaming video, accompanied by a systematic study of navigation memory spanning long-horizon navigation scenarios and real-world validation.*

We propose MEMVLN, a streaming episodic memory framework that endows VLMs with persistent navigation memory. MEMVLN maintains dense short-term observations for immediate action while building a compact, queryable episodic memory over the full navigation history. Concretely, a memory bank progressively stores perceptual features of streaming observations alongside their spatio-temporal information (*i.e.*, odometry and timestep). For action prediction, a memory perceiver extracts latent features from the bank, encodes them with the corresponding spatio-temporal context, and derives memory tokens via cross-attention with learnable queries. The memory tokens are then inserted into the multi-modal sequence within the VLM for auto-regressive action prediction. When the bank reaches its capacity limit, a frame merging strategy consolidates consecutive frames to maintain a coherent episodic summary without discarding history. To further enrich spatial perception, we incorporate a pretrained geometry encoder [18] whose features are fused into the VLM with streaming-optimized efficiency. Both the memory module and the geometry encoder integrate seamlessly into existing VLMs for streaming video without architectural redesign.

To effectively supervise the learning of the memory module, we design a dedicated auxiliary task alongside navigation. Specifically, we adopt trajectory summarization [19, 20, 5] as the supervision for the memory module, where we mask the visual tokens and pass only the memory tokensto the model, forcing it to rely exclusively on episodic memory when generating textutal description of the traversed path. To enrich the supervisory signal, we further decompose each full trajectory into hierarchically structured sub-trajectories [21], yielding fine-grained supervision across multiple temporal scales. Together, the memory module design, multi-granularity trajectory summarization, and masking strategy constitute a principled framework for learning navigation memory.

We conduct extensive experiments in both simulation and real-world environments to validate the efficacy of MEMVLN. On standard VLN-CE benchmarks [1, 22, 2], MEMVLN achieves over 55% success rate on both R2R and RxR, establishing state-of-the-art performance among methods trained on comparable data without massive data scaling, and even surpassing several methods trained on substantially larger datasets. Our ablation studies showcase detailed contributions of different components, such as the memory module and auxiliary supervision. Critically, we underscore the effectiveness of our episodic memory design through long-horizon navigation analyses and physical robot deployment. Real-world evaluation on the G1 robot demonstrates that MEMVLN improves success rate over the VLM baseline from 17% to 40%. To further address practical deployment challenges, we collect 24 real-world demonstrations for post-training, which lifts real-world success rate to 47%, underscoring the adaptability of MEMVLN to real-world navigation.

2 Related Work

Vision-Language Navigation. VLN requires an embodied agent to follow natural language instructions through 3D environments. Early works establish VLN as a foundational task for embodied AI [23, 24, 25, 26, 27, 1, 28, 29, 22, 2, 30, 31, 32, 33]. Given the scarcity of annotated navigation data, considerable efforts have been devoted to data-centric approaches, including data augmentation [34, 35, 36, 21, 37, 38] and data scaling in both simulation environments [39, 40, 41] and the real world [42, 43, 5, 44]. As the model capability advances rapidly, the VLN task has evolved to encompass increasingly complex evaluation scenarios [45, 46, 47, 48, 49, 50, 51]. In this paper, we focus on the fundamental VLN task in continuous environments and study how navigation memory can be learned and leveraged without relying on massive data scaling [1, 22, 2, 21].

Vision-Language Model for Navigation. Effective VLN requires models to perceive 3D environments and comprehend natural language instructions. While early approaches trained navigation models from scratch [52, 53, 54, 39], the strengths of pretrained VLMs have emerged as a more promising pathway towards general-purpose VLN [55, 56, 57, 58, 59, 60, 61]. This paradigm has given rise to a diverse set of approaches, including topological structure methods [62, 63, 64, 65, 66, 67, 13], explicit 3D spatial modeling [68, 69, 70, 71, 72, 73], and trajectory-level modeling [74, 75, 76, 77, 78]. More recently, the input formulation has been simplified to monocular streaming video, under which navigation performance has improved rapidly [19, 20, 4, 5, 6, 16, 17, 79, 7, 80], with significant progress also demonstrated in real-world deployment [81, 82, 83, 84, 85, 86, 87, 88]. Nevertheless, navigation capability under this formulation remains fundamentally constrained by the excessive computational cost of processing streaming video and the absence of navigation memory. Early attempts to address this through implicit recurrent modules exhibit limited capacity [8, 9, 10]. Subsequent works employing explicit memory [11, 12, 14, 15, 89] have yet to achieve optimal performance, suffer from limited compatibility with streaming VLMs, and provide insufficient analyses of memory modules and real-world applicability. In this work, we focus on streaming episodic memory for the VLN task and demonstrate its importance in both simulation and real-world environments.

3 Method

Task Formulation. We study vision-language navigation in continuous environments (VLN-CE), where an embodied agent receives streaming observations and follows language instructions through 3D environments. At each timestep t, the agent receives an egocentric observation o_t and a task instruction $\mathcal{T}$, and predicts an action a_t. The objective is to learn a navigation policy $p(a_t \mid o_{1:t}, \mathcal{T})$. Following prior work [5], actions are represented as text comprising an action type $a_t^{\text{type}} \in \{\texttt{move forward}, \texttt{turn left}, \texttt{turn right}, \texttt{stop}\}$ and an action scale a_t^{scale}, where forward distances are in {25, 50, 75}cm and turn angles are in $\{15°, 30°, 45°\}$.

3.1 Model Overview

As illustrated in Fig. 2, MemVLN comprises four components: the visual encoder and large language model (LLM) of a pretrained VLM, a geometry encoder, and a memory module. At each navigation step, the model receivess visual observation $o_t \in \mathbb{R}^{H\times W\times 3}$ and a text instruction $\mathcal{T}$. Following the standard VLM design, the images o_t are passed into a visual encoder π_V into patch features $f_t \in \mathbb{R}^{H'\times W'\times d}$. We further leverage recent progress on pre-trained visual geometry models [90, 18] to improve the spatial understanding capability of the base VLM. Specifically, we leverage the StreamingVGGT [18] as the geometric encoder and extract the geometric features g_t^{cam} (*i.e.*, camera tokens) from the image sequence. We employ a lightweight geometry adapter ϕ that aligns g_t^{cam} to the VLM feature space and obtain the perceptual features $h_t = f_t + \phi(g_t^{\text{cam}})$. In addition to perceptual features, we leverage image features f_t to generate memory tokens m_t which we detail in Sec. 3.2.

Vision-Language Model Design. MemVLN augments the vanilla LLM input with memory tokens and geometry features, modeling $p(a_t \mid m_t, h_{(t-k):t}, \mathcal{T})$. To enhance awareness of the current situation, we partition the local context into history observations $h_{(t-k):(t-1)}$ and the current observation

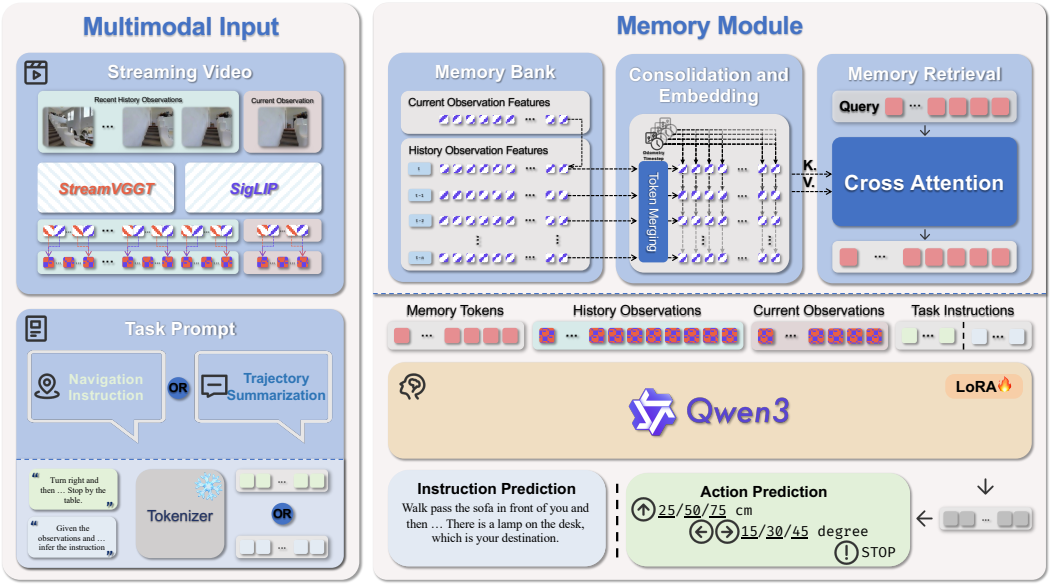


Figure 2: **MEMVLN model overview.** MEMVLN comprises a visual encoder (SigLIP), a geometry encoder (StreamVGGT), a memory module, and an LLM (Qwen3). MEMVLN receives two complementary streams: a short-term context for rich local perception, and an episodic memory for long-horizon reasoning.

h_t. The interleaved multi-modal sequence fed to the LLM π_L is structured as:

$$\Big[\{\underbrace{\mathcal{P}_{\text{mem}}}_{\substack{\text{memory}\\\text{prompt}}}, \underbrace{m_t}_{\substack{\text{memory}\\\text{tokens}}}\}, \{\underbrace{\mathcal{P}_{\text{hist}}}_{\substack{\text{history}\\\text{observation}\\\text{prompt}}}, \underbrace{h_{(t-k):(t-1)}}_{\substack{\text{history}\\\text{observation}\\\text{tokens}}}\}, \{\underbrace{\mathcal{P}_{\text{curr}}}_{\substack{\text{current}\\\text{observation}\\\text{prompt}}}, \underbrace{h_t}_{\substack{\text{current}\\\text{observation}\\\text{tokens}}}\}, \underbrace{\mathcal{T}}_{\substack{\text{task}\\\text{instruction}}}, \underbrace{a_t}_{\substack{\text{action}\\\text{output}}}\Big], \tag{1}$$

$$a_t = \pi_L(m_t, h_{(t-k):t}, \mathcal{T}). \tag{2}$$

The history observation components $\mathcal{P}_{\text{hist}}$ and $h_{(t-k):(t-1)}$ are omitted at the initial step $t = 1$. The overall design endows MEMVLN with comprehensive perception and reasoning capabilities for long-horizon VLN tasks: the dense short-term context anchors fine-grained local perception, while the retrieved memory supplies long-horizon evidence of the agent's full navigational history.

3.2 Memory Module

Memory Bank. Upon receiving a new observation o_t, the corresponding visual feature f_t is appended to an episode-specific memory bank along with its timestep t and odometry z_t. The odometry encodes position (x, y) and orientation $(\cos\theta, \sin\theta)$ relative to the starting point, obtained from onboard sensors or accumulated action execution. When the bank reaches its maximum capacity, the two consecutive frames with the highest average patch-level cosine similarity are merged, preserving a coherent episodic summary without discarding history. We store the visual feature f_t rather than the hybrid feature h_t, as empirical results favor f_t; we attribute this to optimization instability that may arise from incorporating geometry features before the geometry adapter converges.

Memory Perceiver. To derive a compact and efficient memory representation, a memory perceiver resamples the memory bank using a set of learnable queries via cross-attention [91, 92]. Prior to serving as keys and values, the bank features undergo two operations:

- **Spatial Downsampling.** To reduce computational cost, patch features within each frame are spatially merged using ToMe [93]:
$$\bar{f}_{1:t} = \texttt{ToMe}(f_{1:t}). \tag{3}$$
- **Spatio-temporal Embedding.** Timestep and odometry information are encoded into both the bank features and learnable queries q via sinusoid temporal embedding [91] and Fourier spatial

feature [94, 95, 96]:

$$\bar{f}^{+}_{1:t} = \bar{f}_{1:t} + \texttt{sinusoid}(1:t) + \texttt{Fourier}(z_{1:t}), \tag{4}$$

$$q^{+} = q + \texttt{sinusoid}(t) + \texttt{Fourier}(z_t). \tag{5}$$

The memory perceiver then yields a compact and fixed-size set of memory tokens via cross-attention:

$$m_t = \texttt{cross-attn}(q^{+}, \bar{f}^{+}_{1:t}). \tag{6}$$

The memory module offers three key advantages for VLN: (1) **Long-horizon feasibility**: the global memory bank retains local evidence (*e.g.*, signs, doorways, landmarks) throughout the episode, deferring retrieval to when it is needed; in contrast, recurrent memory must immediately decide what to retain at each step. (2) **Computational efficiency**: spatial and temporal consolidation based on feature similarity preserves representative visual cues while maintaining a moderate computational budget. (3) **Informative retrieval**: memory resampling is spatially and temporally grounded through odometry and timestep embeddings, enriching the episodic context available for decision-making.

3.3 Data and Training

MemVLN is trained jointly on two tasks: VLN and trajectory summarization. The navigation task teaches the model to predict the next action, while the summarization task shapes the memory representation towards route-level semantics. Both tasks are trained with a unified cross-entropy loss.

Vision-Language Navigation. VLN trajectories are preprocessed with action merging following NaVILA [5], where consecutive atomic actions within a window of three steps are merged, *e.g.*, three consecutive "`move forward 25cm`" actions become a single "`move forward 75cm`". This enriches action diversity and improves training efficiency. At each step, the model is supervised to predict the ground-truth action a^{*}_t:

$$\mathcal{L} = -\log p(a^{*}_t \mid m_t, h_{(t-k):t}, \mathcal{T}). \tag{7}$$

Trajectory Summarization. The trajectory summarization task provides auxiliary supervision for the memory module. Using the same VLN trajectories, the task instruction $\mathcal{T}$ is repurposed as the ground-truth label. Crucially, all visual observation tokens $h_{(t-k):t}$ are masked (invisible) to the LLM, leaving only the memory tokens m_t as visual evidence. This masking strategy forces the model to reconstruct navigational semantics exclusively from episodic memory, supervising it to retain evidence relevant to long-horizon instruction following, such as landmarks marking completed subgoals or visual cues indicating stopping positions. Let $\mathcal{T}_{(\mathrm{TS})}$ denote the task-specific prompt for trajectory summarization, the loss can be expressed as:

$$\mathcal{L} = -\log p(\mathcal{T} \mid m_t, \mathcal{T}_{(\mathrm{TS})}). \tag{8}$$

Sub-trajectory Derivation. To further enrich supervision granularity, we augment the whole-trajectory summarization data with sub-trajectory data. Starting from the FGR2R annotations [21], which provide sub-instructions and corresponding waypoints in discrete navigation graphs, we calibrate these waypoints to continuous environments by retrieving the nearest points on the continuous trajectory. This yields sub-trajectory splits that provide finer-grained supervision, encouraging the memory module to capture semantics at multiple temporal scales.

4 Experiment

Implementation Details. We adopt Qwen3-VL-4B [103] as the VLM backbone, and StreamVGGT-1B [18] as the geometry encoder. We freeze the visual encoder and geometry encoder, finetune the LLM with LoRA [104], and train the memory module and geometry adapter from scratch, yielding 1B learnable parameters in total. The local window size k is set to 16. The model is trained for 1 epoch (2 days on 8×A100 80G GPUs) using the AdamW optimizer [105] with a base learning rate of 3×10^{-5}, scheduled with linear warmup and cosine decay. See detailed hyperparameters in *Appendix*.

Table 1: **Evaluation results on VLN-CE R2R Val-Unseen split.** "Pano.", "Odo.", and "S.RGB" represent panoramic view, odometry, and single RGB. Note that odometry in MemVLN is derived from accumulated action execution rather than sensor input. External data refers to training sources beyond the standard R2R/RxR datasets (*e.g.*, EnvDrop [34]). The results are partially referenced from JanusVLN [16]. * denotes baselines with a comparable training data configuration.

Method	Observation				R2R Val-Unseen				Training
	Pano.	Odo.	Depth	S.RGB	NE↓	OS↑	SR↑	SPL↑	External Data
CMA [64] [CVPR22]	✓	✓	✓		6.20	52.0	41.0	36.0	-
VLN↻BERT [10] [CVPR21]	✓	✓	✓		5.74	53.0	44.0	39.0	-
Ego^2-Map [97] [ICCV23]	✓	✓	✓		5.54	56.0	47.0	41.0	-
DreamWalker [98] [ICCV23]	✓	✓	✓		5.53	59.0	49.0	44.0	-
AO-Planner [99] [AAAI25]	✓		✓		5.55	59.0	47.0	33.0	-
g3D-LF [100] [CVPR25]		✓	✓	✓	5.70	59.5	47.2	34.6	-
NaVid-4D [101] [ICRA25]			✓	✓	5.99	55.7	43.8	37.1	-
NavMorph [102] [ICCV25]			✓	✓	5.75	56.9	47.9	33.2	-
NaVid [20] [RSS24]				✓	5.47	49.1	37.4	35.9	953K
Sim2Real [69] [CoRL24]				✓	5.95	55.8	44.9	30.4	0K
StreamVLN* [6] [ICRA26]				✓	6.05	53.8	45.5	41.6	10033K
Uni-NaVid [4] [RSS25]				✓	5.58	53.3	47.0	42.7	3577K
NaVILA* [5] [RSS25]				✓	5.37	57.6	49.7	45.5	12574K
JanusVLN* [16] [ICLR26]				✓	**5.17**	58.0	52.8	49.2	0K
MemVLN (Ours)				✓	5.19	**61.2**	**55.0**	**50.5**	0K

Data and Evaluation. We train on the official training splits of R2R [1] and RxR [22], comprising 10,819 and 19,996 demonstrations, respectively. The trajectory summarization data is derived from these same demonstrations. No additional navigation datasets or general multi-modal data are incorporated. For evaluation in simulation, we roll out the policy in Habitat-Sim [27] following the VLN-CE protocol [2], and report results on the Val-Unseen split. For R2R, we report success rate (SR), oracle success rate (OSR), success weighted by path length (SPL), and navigation error (NE). For RxR, we report SR, SPL, normalized dynamic time warping (nDTW) [106], and NE.

4.1 Main Simulation Results

Comparison Baselines. We compare primarily against recent VLM-based VLN methods, with particular focus on those adopting a streaming video formulation, including StreamVLN [6] and JanusVLN [16]. For methods reporting results under multiple data configurations, we select the configuration closest to our R2R/RxR-only setting. External training data information is noted in Tab. 1, referenced from JanusVLN. Specifically, we compare against StreamVLN* (add EnvDrop data), NaVILA* (add EnvDrop and VQA data), and JanusVLN* (without external data).

Results. As shown in Tabs. 1 and 2, MemVLN achieves state-of-the-art performance on most metrics among methods trained under comparable data configurations, with the sole exception of R2R NE where it performs on par with JanusVLN*. The results show that the streaming video formulation ("S.RGB") has emerged as a compelling direction, combining strong performance with a streamlined and deployment-friendly input interface. Within this category, MemVLN outperforms all methods without external training data, underscoring the benefits of episodic memory and the improved training efficiency of our approach. Notably, the performance gain is more pronounced on RxR, which features longer instructions and more complex navigation trajectories, demonstrating the particular efficacy of episodic memory in long-horizon navigation scenarios.

4.2 Ablation Studies

Settings. Sarting from a naive VLM baseline, we progressively incorporate three components: the memory module (mem.), geometry encoder (geo.), and auxiliary supervision from trajectory summarization (aux.). The combination of all three constitutes the default MemVLN. We report evaluation results on both R2R-CE and RxR-CE. See more ablations on hyperparameters in *Appendix*.

Table 2: **Evaluation results on VLN-CE RxR Val-Unseen split.** Settings follow Tab. 1 and JanusVLN [16].

Method	Observation				RxR Val-Unseen				Training
	Pano.	Odo.	Depth	S.RGB	NE↓	SR↑	SPL↑	nDTW↑	External Data
CMA [64] [CVPR22]	✓	✓	✓		8.76	26.5	22.1	47.0	-
VLN↻BERT [10] [CVPR21]	✓	✓	✓		8.98	27.0	22.6	46.7	-
AO-Planner [99] [AAAI25]	✓		✓		7.06	43.3	30.5	50.1	-
NavMorph [102] [ICCV25]			✓	✓	8.85	30.8	22.8	44.2	-
Sim2Real [69] [CoRL24]				✓	8.79	36.7	25.5	18.1	0K
Uni-NaVid [4] [RSS25]				✓	6.24	48.7	40.9	-	3577K
NaViLA [5] [RSS25]				✓	6.77	49.3	44.0	58.8	13132K
JanusVLN* [16] [ICLR26]				✓	6.46	51.4	44.3	59.1	0K
StreamVLN [6] [ICRA26]				✓	6.22	52.9	46.0	61.9	~26330K
MemVLN (Ours)				✓	**5.59**	**55.4**	**48.4**	**64.6**	0K

Table 3: **Ablation studies on memory module (mem.), geometry encoder (geo.), and auxiliary task (aux.).**

Setting	R2R Val-Unseen				RxR Val-Unseen			
	NE↓	OS↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
Baseline	6.56	51.1	43.9	40.9	6.90	48.8	43.9	59.7
+ mem.	6.38	56.7	45.6	41.6	6.28	53.3	45.9	59.6
+ mem. + geo.	5.85	56.3	49.7	45.8	5.74	**56.1**	**49.4**	63.0
+ mem. + geo. + aux.	**5.19**	**61.2**	**55.0**	**50.5**	**5.59**	55.4	48.4	**64.6**

Results. As shown in Tab. 3, all three components yield consistent performance improvements. Adding the memory module considerably improves R2R OSR and RxR SR, demonstrating the benefit of retaining an episodic memory. Incorporating the geometry encoder significantly improves all metrics, confirming that enhanced spatial understanding translates directly to stronger navigation capability. Finally, the auxiliary supervision further improves performance, with a more pronounced gain on R2R. This suggests that multi-granularity sub-trajectory supervision effectively shapes the memory module towards fine-grained comprehension of navigation behaviors.

4.3 Long-Horizon Analysis

Settings. We conduct long-horizon analysis on the RxR Val-Unseen split, which features complex instructions and long-horizon navigation episodes. Task complexity is quantified by the number of clauses in each instruction, delimited by "," and ".", and categorized as: simple (1-3), medium (4-6), complex (7-10), and very complex (>10). We evaluate three model variants: the VLM baseline, baseline with memory module, and the full MemVLN. Beyond task complexity, we further categorize episodes by whether the instruction involves: (i) backtracking, (ii) clear stop signs and long hallways, and (iii) passing landmarks. We report the success rate (SR) and navigation error (NE) metrics.

Results. The results in Fig. 3 show that across all categories, "baseline < +Mem < Full" is consistent, confirming the contribution of the memory module and auxiliary supervision. In the complexity analysis, the benefit of the memory module grows more pronounced as instruction complexity increases. In the instruction-type analysis, subsets involving backtracking, stop signs, and passing landmarks exhibit wider performance gaps between models with and without the memory module, underscoring the importance of episodic memory in demanding navigation scenarios. These findings demonstrate that MemVLN is particularly effective in long-horizon and complex navigation settings.

4.4 Real-world Experiment

Settings. We deploy MemVLN on a real-world G1 robot. We devise 30 instructions and let the policy execute the instructions with human interventions to collect demonstrations, recording both model predictions and the corrected actions. We hold out 6 demonstrations for evaluation, with the remaining 24 used for post-training. Post-training adopts behavior cloning with step-wise loss

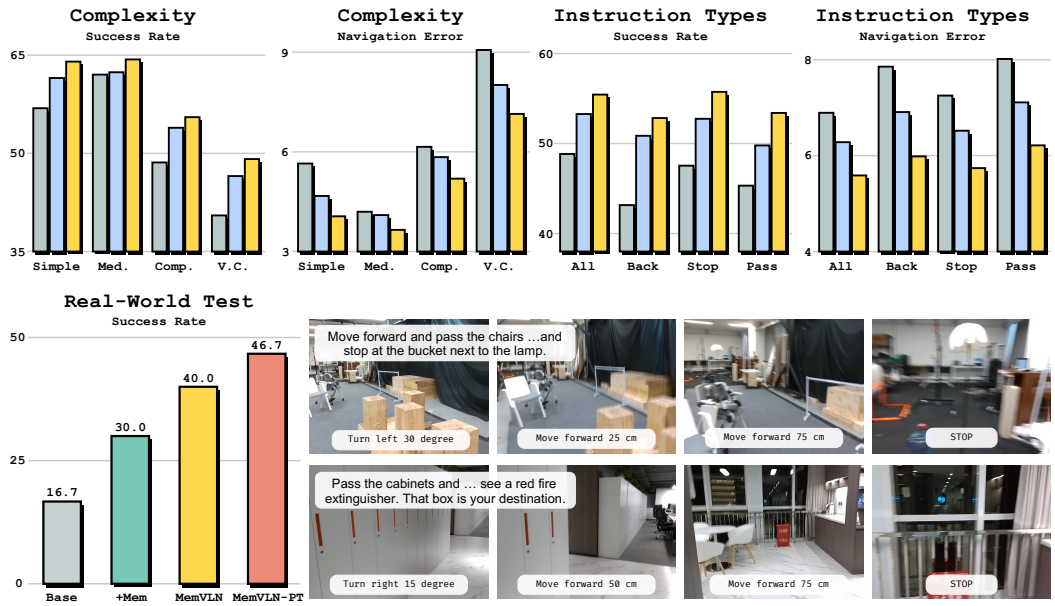


Figure 3: **Long-horizon and real-world analysis**. Complexity denotes the number of clauses in task instruction: simple (1-3), medium (4-6), complex (7-10), and very complex (>10). Instruction types indicate whether the instruction involves (i) backtracking, (ii) clear stop points and long hallways, or (iii) passing landmarks. The bottom row illustrates the real-world success rate and qualitative results of real-world deployment.

weighting, assigning larger weights to human-corrected steps. We evaluate four model variants: the VLM baseline, baseline with memory module, the full MEMVLN, and MEMVLN after post-training (PT). Each instruction is evaluated over 5 independent runs, and we report the average success rate.

Results. Real-world results are presented in Fig. 3, showing consistent improvements from adding the memory module, auxiliary supervision, and real-world post-training. Qualitative examples further illustrate navigation episodes involving challenging conditions such as long hallways, cluttered scenes, and large turning maneuvers. MEMVLN handles these scenarios effectively, and post-training yields further gains, demonstrating the adaptability of MEMVLN to real-world navigation.

5 Conclusion

We present MEMVLN, a streaming episodic memory framework for vision-language navigation that addresses the fundamental memory bottleneck of VLM-based streaming navigation models. MEMVLN introduces a plug-and-play memory module that maintains a coherent episodic summary of the full navigation history, complemented by a pretrained geometry encoder for enriched spatial perception. To effectively train the memory module, we design a masking-based trajectory summarization task with multi-granularity supervision, encouraging the model to reason exclusively from episodic memory. Extensive experiments on VLN-CE benchmarks demonstrate state-of-the-art performance without massive data scaling, and both simulation and real-world analyses validate the efficacy of episodic memory. Real-world post-training with a small number of demonstrations further improves real-world performance, demonstrating the practical adaptability of MEMVLN.

6 Limitations and Future Work

First, memory consolidation is primarily similarity-based and may inadvertently discard task-relevant observations in excessively long-horizon episodes. Second, while trajectory summarization effectively supervises memory learning, broader training strategies remain under-explored [107]. Finally, while real-world post-training yields notable gains, practical deployment challenges such as collision behaviors persist. Future work will investigate advanced strategies in memory consolidation and memory learning, and more effective real-world post-training such as reinforcement learning (RL).

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